

# ANALOG ELECTRONICS (AE)

Exam Crash Revision • Predicted Questions • Important Topics

**Exam pattern:** 5 questions × 10 marks = 50 marks • 1 question from each module • 20–25 marks expected from numericals

## ■ HIGH-PROBABILITY PREDICTED PAPER

|               |                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|
| Q1 — Module 1 | Multistage amplifier: DC + mid-frequency analysis, voltage gain, and/or lower & upper cutoff frequencies   |
| Q2 — Module 2 | Current mirror/current source: derive output current OR solve a numerical. Wilson/Widlar may appear as     |
| Q3 — Module 3 | Op-amp applications: derive/design inverting and non-inverting amplifiers. Adder/subtractor/comparator a   |
| Q4 — Module 4 | Power amplifiers: Class A numerical OR Class B/AB efficiency derivation, with comparison, waveforms a      |
| Q5 — Module 5 | RC phase-shift oscillator or crystal oscillator. Barkhausen criterion and oscillator theory are important. |

## ■ MODULE 1 — FREQUENCY RESPONSE & MULTISTAGE AMPLIFIERS

**Sir's focus:** Multistage amplifiers; DC + mid-frequency analysis;  $f_L$  and  $f_H$  parameters.

### 1. Multistage Amplifier

A multistage amplifier connects two or more amplifier stages in cascade. The output of one stage becomes the input of the next. The main reason for cascading is to obtain higher overall voltage gain.

**Key relation:**  $A_v(\text{total}) = A_{v1} \times A_{v2} \times \dots \times A_{vn}$

**Remember:** Gain increases with cascading, while the overall bandwidth generally becomes narrower.

### 2. DC Analysis

Find the DC operating point (Q-point): collector/drain current and voltage. DC analysis normally treats coupling and bypass capacitors as open circuits.

For an exam answer: draw the circuit → replace capacitors appropriately → calculate currents/voltages → state the operating point.

### 3. Mid-Frequency Analysis

At mid frequencies, coupling and bypass capacitors are approximately short circuits and transistor parasitic capacitances are neglected. The amplifier therefore operates approximately at its maximum/constant gain.

### 4. Lower and Upper Cutoff Frequencies

**$f_L$ :** Lower cutoff frequency. It is mainly associated with coupling and bypass capacitors and the low-frequency roll-off.

**$f_H$ :** Upper cutoff frequency. It is mainly associated with transistor/internal capacitances and the high-frequency roll-off.

**Bandwidth:**  $BW = f_H - f_L$

**Single-pole RC cutoff:**  $f = 1/(2\pi RC)$

**Exam diagram:** Draw the frequency-response curve showing the midband gain,  $f_L$ ,  $f_H$  and the  $-3$  dB points.

## 5. Miller Effect — Backup Topic

An amplifier's feedback capacitance between input and output can appear as a larger effective input capacitance because of voltage gain. This increases the high-frequency limitation and can reduce bandwidth.

## ■ MODULE 2 — CURRENT MIRRORS & CURRENT SOURCES

**Sir's focus:** Derive output current OR solve numerical problems.

### 1. Basic Current Mirror

A current mirror generates an output current that approximately copies a reference current. It is widely used for biasing in integrated circuits.

**Ideal approximation:**  $I_{out} \approx I_{ref}$

For a BJT mirror, matched transistors have approximately equal  $V_{BE}$ , so their collector currents are approximately equal when operating in the active region. Base currents introduce error.

### 2. What to Write in a Derivation

Draw the current-mirror circuit → identify  $I_{ref}$  and  $I_{out}$  → use the matched-device condition → write transistor current equations → obtain the  $I_{out}$  relation → state assumptions.

### 3. Wilson Current Source — Backup

Wilson current sources use additional transistor feedback to improve current-copying accuracy and increase output resistance compared with a simple current mirror.

### 4. Widlar Current Source — Backup

The Widlar current source modifies a basic current mirror using an emitter resistor (for BJT implementation) to generate small output currents without requiring a very large resistor elsewhere.

## ■ MODULE 3 — OP-AMP APPLICATIONS

**Sir's focus:** Inverting and non-inverting amplifiers + applications.

### 1. Ideal Op-Amp Rules

For an ideal op-amp with negative feedback: input currents are approximately zero and  $V_+ \approx V_-$  (virtual short).

### 2. Inverting Amplifier — MUST KNOW

Input is applied through  $R_{in}$  to the inverting terminal; the non-inverting terminal is grounded. The output is  $180^\circ$  out of phase with the input.

**Gain:**  $A_v = V_o/V_i = -R_f/R_{in}$

Derivation: use the virtual-ground condition at the inverting node and apply KCL. The input current through  $R_{in}$  equals the feedback current through  $R_f$ .

### 3. Non-Inverting Amplifier — MUST KNOW

The input is applied to the non-inverting terminal. Output is in phase with the input.

**Gain:**  $A_v = 1 + R_f/R_1$

### 4. Other Op-Amp Applications — BACKUP / PYQ AREA

Adder (summing amplifier), subtractor/difference amplifier, comparator, sign changer, scale changer and Schmitt trigger. For a design question: identify the required equation → select the configuration → write the gain equation → choose resistor ratios → verify  $V_o$ .

## ■ MODULE 4 — POWER AMPLIFIERS

**Sir's focus:** Class A, B, AB, C; comparison; waveforms; collector current; Class A numericals; Class B & AB efficiency derivation.

| Class | Conduction angle | Main feature                                                 | Ideal maximum efficiency                     |
|-------|------------------|--------------------------------------------------------------|----------------------------------------------|
| A     | 360°             | Low distortion; conducts for full cycle                      | 25% (RC-coupled) / 50% (transformer-coupled) |
| B     | 180°             | Each device conducts for half cycle; push-pull commonly used | 78.5%                                        |
| AB    | 180°–360°        | Reduces crossover distortion compared with B                 | Between A and B                              |
| C     | <180°            | Highly efficient; used with tuned circuits/RF applications   | High                                         |

### 1. Class A Numerical — MUST PRACTICE

Typical data may include VCC, quiescent current, load resistance and signal current/voltage. Be ready to calculate DC input power, AC output power, efficiency and transistor power dissipation.

**Basic relations:**  $P_{DC} = V_{CC} \times I_{DC}$ ;  $\text{efficiency } \eta = P_{AC}(\text{out})/P_{DC} \times 100\%$

### 2. Class B Efficiency — MUST KNOW

For an ideal push-pull Class B amplifier, the maximum theoretical efficiency is:

$$\eta_{\max} = \pi/4 = 78.5\%$$

Know the derivation using sinusoidal output current/voltage, DC input power and AC output power. Also know why the practical efficiency is lower.

### 3. Class AB

Class AB biases the output devices slightly above cutoff so each device conducts for more than 180° but less than 360°. Its main purpose is to reduce crossover distortion while retaining better efficiency than Class A.

### 4. Waveforms + Collector Current

Be able to sketch the input/output sinusoid and the collector/drain current conduction interval for Classes A, B, AB and C. Label the conduction angle.

## ■ MODULE 5 — OSCILLATORS

**Sir's focus:** Mostly theoretical — RC phase-shift oscillator and crystal oscillator.

### 1. Barkhausen Criterion — MUST REMEMBER

For sustained oscillations, the loop gain condition is:

$$|A\beta| = 1 \text{ and } \angle A\beta = 0^\circ \text{ (or } 360^\circ)$$

The first condition gives the required loop-gain magnitude; the second ensures positive feedback at the oscillation frequency.

### 2. RC Phase-Shift Oscillator — MUST DO

It uses an amplifier and an RC feedback network. Three RC sections are commonly used to provide approximately  $180^\circ$  phase shift, while the amplifier provides another  $180^\circ$ .

**For three identical RC sections:**  $f \approx 1/(2\pi RC\sqrt{6})$

Also remember the practical gain condition associated with the RC network, the circuit diagram, working, and frequency expression.

### 3. Crystal Oscillator — MUST DO

A crystal oscillator uses the piezoelectric resonance of a quartz crystal to obtain a highly stable oscillation frequency. It is used where frequency stability is important, such as clocks and timing circuits.

For a 10-mark answer: draw the circuit → explain piezoelectric effect/resonance → explain feedback and oscillation → state frequency stability advantage → mention applications.

## ■ LAST-NIGHT PRIORITY LIST

|   |          |                                                                             |
|---|----------|-----------------------------------------------------------------------------|
| 1 | Module 1 | Multistage amplifier + DC/mid-frequency analysis + fL/fH                    |
| 2 | Module 3 | Inverting & non-inverting op-amp derivations + applications                 |
| 3 | Module 4 | Class A numerical + Class B efficiency derivation + A/B/AB/C comparison     |
| 4 | Module 5 | RC phase-shift oscillator + crystal oscillator + Barkhausen criterion       |
| 5 | Module 2 | Current mirror output-current derivation/numerical; Wilson/Widlar as backup |

## ■ 10-MARK ANSWER TEMPLATE

For derivation/theory questions: **(1)** Definition/intro → **(2)** neat circuit diagram → **(3)** working/principle → **(4)** equations/derivation → **(5)** final formula/result boxed → **(6)** waveform/graph if applicable → **(7)** applications/advantages.

For numericals: **Given** → **Required** → **Formula** → **Substitution** → **Calculation** → **Final answer with unit.**

## ■■ EXAM REMINDER

Your sir specifically said: **“Solve PYQs → guaranteed questions.”** Therefore, after revising these topics, spend the remaining time solving the shared PYQs rather than starting unrelated topics.